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S. J. C. INSTITUTE OF TECHNOLOGY, CHICKBALLAPUR

Department of Mathematics

LECTURE NOTES

Advanced Calculus and Numerical Methods (21MAT21)

Module-1 (Integral Calculus)

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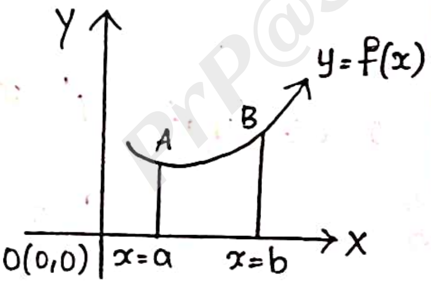
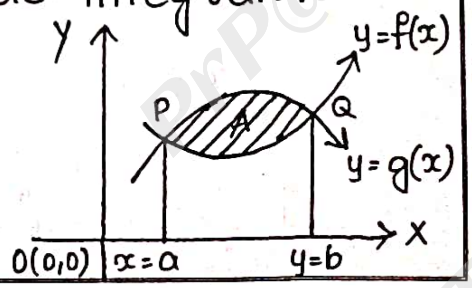
Module-01 → Integral Calculas

Some important results:-

1. Suppose $y=f(x)$ be a real value function defined over the region 'R', bounded by the lines $x=a, x=b$, then it can be represented by as $\int_{x=a}^{b} f(x)dx$ which will give the length of the curve from $x=a$ to $x=b$, which is also called the "Line integrals."



2. Suppose $u=\phi(x,y)$ be a real value function defined over the region 'R', bounded by the lines $x=a, x=b$ and the curves $y=f(x)$ and then it can be represented as $\int_{x=a}^{b}\int_{y=f(x)}^{g(x)} \phi(x,y)dy\,dx$ which is called "Double integrals". and also which will give the area of the bounded region 'R'.



3. Suppose $u = \phi(x,y,z)$ is the function is defined in the space 'S', bounded by $x=a$, $x=b$, $y=f_1(x)$, $y=f_2(x)$, $z=g_1(x,y)$, $z=g_2(x,y)$, then it can be represented as

$$\int_{x=a}^{b}\int_{y=f_1(x)}^{f_2(x)}\int_{z=g_1(x,y)}^{g_2(x,y)} \phi(x,y,z)\,dz\,dy\,dx$$

, which is called the "Triple integral" and which is also used to find the volume of a sphere, cone, tetrahedral.

4. $$\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = \begin{cases} \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdots\frac{1}{2}\,\frac{\pi}{2} & n \text{ is even} \\ \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdots\frac{2}{3}\cdot 1 & n \text{ is odd.} \end{cases}$$

Example:-

1. $$\int_0^{\pi/2} \sin^3 x\,dx = \frac{3-1}{3} = \frac{2}{3}$$

2. $$\int_0^{\pi/2} \cos^4 x\,dx = \frac{4-1}{4}\cdot\frac{4-3}{4-2}\cdot\frac{\pi}{2} = \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{3\pi}{16}$$

1. Evaluate $\int_0^1\int_0^x (x^2+y^2)\,dy\,dx$.

Sol:- Let

$$I = \int_0^1\int_0^x (x^2+y^2)\,dy\,dx.$$

$$I = \int_{x=0}^{1}\int_{y=0}^{x} (x^2+y^2)\,dy\,dx$$

$$I = \int_{x=0}^{1}\left|\int_{y=0}^{x} (x^2+y^2)\,dy\right| dx$$

$$I = \int_{x=0}^{1}\left\{x^2\int_{y=0}^{x} 1\,dy + \int_{y=0}^{x} y^2\,dy\right\}dx$$

$$I = \int_{x=0}^{1}\left\{x^2\left[y\right]_0^x + \left[\frac{y^3}{3}\right]_0^x\right\}dx$$

$$I = \int_{x=0}^{1}\left|x^2(x) + \frac{x^3}{3}\right|dx$$

$$I = \int_{x=0}^{1}\left(x^3 + \frac{x^3}{3}\right)dx$$

$$I = \int_{x=0}^{1}\frac{4}{3}x^3\,dx$$

$$I = \frac{4}{3}\int_{x=0}^{1}x^3\,dx$$

$$= \frac{\cancel{4}}{3}\left|\frac{x^4}{\cancel{4}}\right|_0^1$$

$$\boxed{I = \frac{1}{3}}$$

2. Evaluate $\int_0^1\int_0^1 (x+y)\,dy\,dx$.

Sol:- Let $I = \int_0^1\int_0^1 (x+y)\,dy\,dx$

$$I = \int_{x=0}^{1}\int_{y=0}^{1}(x+y)\,dy\,dx$$

$$I = \int_{x=0}^{1} \left| \int_{y=0}^{1} (x+y)\,dy \right| dx$$

$$I = \int_{x=0}^{1} \left| x \int_{y=0}^{1} 1\,dy + \int_{y=0}^{1} y\,dy \right| dx$$

$$I = \int_{x=0}^{1} \left| x\,[y]_0^1 + \left[\frac{y^2}{2}\right]_0^1 \right| dx$$

$$I = \left| \frac{x^2}{2} + \frac{x}{2} \right|_0^1$$

$$I = \frac{1}{2} + \frac{1}{2}$$

$$\boxed{I = 1}$$

3. Evaluate $\int_{-c}^{c}\int_{-b}^{b}\int_{-a}^{a}(x^2+y^2+z^2)\,dz\,dy\,dx$.

Sol:- Let $I = \int_{x=-c}^{c}\int_{y=-b}^{b}\int_{z=-a}^{a}(x^2+y^2+z^2)\,dz\,dy\,dx$.

$$I = \int_{x=-c}^{c}\int_{y=-b}^{b}\left[\int_{z=-a}^{a}(x^2+y^2+z^2)\,dz\right]dy\,dx$$

$$I = \int_{x=-c}^{c}\int_{y=-b}^{b}\left[x^2z+y^2z+\frac{z^3}{3}\right]_{-a}^{a}dy\,dx$$

$$I = \int_{x=-c}^{c}\int_{y=-b}^{b}\left\{\left[ax^2+ay^2+\frac{a^3}{3}\right]-\left[-ax^2-ay^2-\frac{a^3}{3}\right]\right\}dy\,dx$$

$$I = \int_{x=-c}^{c}\int_{y=-b}^{b}\left[2ax^2+2ay^2+\frac{2a^3}{3}\right]dy\,dx$$

$$I = 2a\int_{x=-c}^{c}\left[\int_{y=-b}^{b}\left(x^2+y^2+\frac{a^2}{3}\right)dy\right]dx$$

$$I = 2a\int_{x=-c}^{c}\left[x^2y+\frac{y^3}{3}+\frac{a^2y}{3}\right]_{-b}^{b}dx$$

$$I = 2a\int_{x=-c}^{c}\left[\left(bx^2+\frac{b^3}{3}+\frac{a^2b}{3}\right)-\left(-bx^2-\frac{b^3}{3}-\frac{a^2b}{3}\right)\right]dx$$

$$I = 2a\int_{x=-c}^{c}\left[2bx^2+\frac{2b^3}{3}+\frac{2a^2b}{3}\right]dx$$

$$I = 4ab\int_{x=-c}^{c}\left[x^2+\frac{b^2}{3}+\frac{a^2}{3}\right]dx$$

$$I = 4ab\left|\frac{x^3}{3} + \frac{b^2x}{3} + \frac{a^2x}{3}\right|_{-c}^{c}$$

$$I = 4ab\left|\left(\frac{c^3}{3} + \frac{b^2c}{3} + \frac{a^2c}{3}\right) - \left(-\frac{c^3}{3} - \frac{b^2c}{3} - \frac{a^2c}{3}\right)\right|$$

$$I = 4ab\left|\frac{2c^3}{3} + \frac{2b^2c}{3} + \frac{2a^2c}{3}\right|$$

$$I = \frac{8abc}{3}\left|a^2 + b^2 + c^2\right|$$

4. Evaluate $\int_{-1}^{1}\int_{0}^{z}\int_{x-z}^{x+z}(x+y+z)\,dy\,dx\,dz$

Sol:- Let $I = \int_{z=-1}^{1}\int_{x=0}^{z}\int_{y=x-z}^{x+z}(x+y+z)\,dy\,dx\,dz$

$$I = \int_{z=1}^{1}\int_{x=0}^{z}\left|xy + \frac{y^2}{2} + yz\right|_{x-z}^{x+z} dx\,dz$$

$$I = \int_{z=-1}^{1}\int_{x=0}^{z}\left\{\left|x(x+z) + \frac{(x+z)^2}{2} + z(x+z)\right| - \left|x(x-z) + \frac{(x-z)^2}{2} + z(x-z)\right|\right\} dx\,dz$$

$$I = \int_{x=-1}^{1}\int_{x=0}^{z}\left|x(\cancel{x}+z-\cancel{x}+z) + \frac{1}{2}\left|(x+z)^2 - (x-z)^2\right| + z(\cancel{x}+z-\cancel{x}+z)\right| dx\,dz$$

$$I = \int_{z=-1}^{1}\int_{x=0}^{z}\left(2zx + \frac{1}{\cancel{2}}(\overset{2}{\cancel{4}}zx) + 2z^2\right) dx\,dz$$

$$I = \int_{z=-1}^{1}\int_{x=0}^{z}(2zx + 2zx + 2z^2)\,dx\,dz$$

$$I = \int_{z=-1}^{1}\int_{x=0}^{z}(4zx + 2z^2)\,dx\,dz$$

$$I = \int_{z=-1}^{1}\left|\cancel{4}^{2}\frac{zx^2}{\cancel{2}} + 2z^2x\right|_0^z dz$$

$$I = \int_{z=-1}^{1}(2z^3 + 2z^3)\,dz$$

$$I = \int_{z=-1}^{1}4z^3.\,dz$$

$$I = 4\int_{z=-1}^{1}z^3.\,dz$$

$$I = \cancel{4}\left|\frac{z}{\cancel{4}}\right|_{-1}^{1}$$

$$I = \left|z^4\right|_{-1}^{1}$$

$$I = 1 - 1 = 0$$

$$\boxed{I = 0}$$

5. Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} xyz \; dz\,dy\,dx$.

Sol:- Let $I = \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \int_{z=0}^{\sqrt{a^2-x^2-y^2}} xyz \; dz\,dy\,dx$

$$I = \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} xy \left| \frac{z^2}{2} \right|_{z=0}^{\sqrt{a^2-x^2-y^2}} dy\,dx$$

$$I = \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \frac{xy}{2}(a^2-x^2-y^2)\,dy\,dx$$

$$I = \frac{1}{2} \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \left| a^2xy - x^3y - xy^3 \right| dy\,dx$$

$$I = \frac{1}{2} \int_{x=0}^{a} \left\{ \frac{a^2xy^2}{2} - \frac{x^3y^2}{2} - \frac{xy^4}{4} \right\}_{y=0}^{\sqrt{a^2-x^2}} dx$$

$$I = \frac{1}{2} \int_{x=0}^{a} \left| \frac{a^2x}{2}(a^2-x^2) - \frac{x^3}{2}(a^2-x^2) - \frac{x}{4}(a^2-x^2)^2 \right| dx$$

$$I = \frac{1}{2} \int_{x=0}^{a} \frac{(a^2-x^2)}{2} \left| a^2x - x^3 - \frac{x}{2}(a^2-x^2) \right| dx$$

$$I = \frac{1}{4} \int_{x=0}^{a} (a^2-x^2) \left| \frac{2a^2x - 2x^3 - x(a^2-x^2)}{2} \right| dx$$

$$I = \frac{1}{8} \int_{x=0}^{a} (a^2-x^2)(a^2x - x^3)\,dx$$

$$I = \frac{1}{8}\int_{x=0}^{a}\left[a^4x - a^2x^3 - a^2x^3 + x^5\right]dx$$

$$I = \frac{1}{8}\int_{x=0}^{a}(a^4x - 2a^2x^3 + x^5)dx$$

$$I = \frac{1}{8}\left| a^4\frac{x^2}{2} - \cancel{2}^{1} a^2\frac{x^4}{\cancel{4}_2} + \frac{x^6}{6}\right|_0^a$$

$$I = \frac{1}{8}\left|\cancel{\frac{a^6}{2}} - \cancel{\frac{a^6}{2}} + \frac{a^6}{6}\right|$$

$$I = \frac{1}{8}\left|\frac{a^6}{6}\right|$$

$$I = \frac{a^6}{48}$$

6. Evaluate $\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} xyz\ dz\,dy\,dx.$

Sol:- Let $I = \int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} xyz\ dz\,dy\,dx.$

$$I = \int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}} xy\left(\frac{z^2}{2}\right)_{z=0}^{\sqrt{1-x^2-y^2}} dy\,dx$$

$$I = \frac{1}{2}\int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}}(xy(1-x^2-y^2))\,dy\,dx$$

$$I = \frac{1}{2}\int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}}(xy - x^3y - xy^3)\,dy\,dx$$

$$I = \frac{1}{2}\int_{x=0}^{1}\left(\frac{xy^2}{2} - \frac{x^3y^2}{2} - \frac{xy^4}{4}\right)_{y=0}^{\sqrt{1-x^2}} dx$$

$$I = \frac{1}{4}\int_{x=0}^{1}\left(x(1-x^2) - x^3(1-x^2) - \frac{x(1-x^2)^2}{2}\right) dx$$

$$I = \frac{1}{4}\int_{x=0}^{1}(1-x^2)\left(\frac{2x-2x^3-x+x^3}{2}\right) dx$$

$$I = \frac{1}{8}\int_{x=0}^{1}(1-x^2)(x-x^3)\,dx$$

$$I = \frac{1}{8}\int_{x=0}^{1}(x-x^3-x^3+x^5)\,dx$$

$$I = \frac{1}{8}\int_{x=0}^{1}(x-2x^3+x^5)\,dx$$

$$I = \frac{1}{8}\left(\frac{x^6}{6} - \frac{2x^4}{4} + \frac{x^2}{2}\right)_0^1$$

$$I = \frac{1}{8}\left(\frac{1}{6} - \frac{\cancel{2}}{\cancel{4}_2} + \frac{1}{2}\right)$$

$$I = \frac{1}{8}\left(\frac{1}{6} - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}}\right)$$

$$I = \frac{1}{48}$$

7. Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} \frac{dz\,dy\,dx}{\sqrt{a^2-x^2-y^2-z^2}}$

Sol:- Let $I = \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} \frac{dz\,dy\,dx}{\sqrt{a^2-x^2-y^2-z^2}}$

$$I = \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \int_{z=0}^{\sqrt{a^2-x^2-y^2}} \frac{1}{\sqrt{\left(\sqrt{a^2-x^2-y^2}\right)^2 - z^2}}\, dz\,dy\,dx$$

Let $K = \sqrt{a^2-x^2-y^2}$

$$I = \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \int_{z=0}^{K} \frac{1}{\sqrt{K^2-z^2}}\, dz\,dy\,dx \quad \left(\because \int \frac{1}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)\right)$$

$$I = \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \left|\sin^{-1}\left(\frac{z}{K}\right)\right|_{z=0}^{K} dy\,dx$$

$$I = \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \left|\sin^{-1}(1) - \sin^{-1}(0)\right| dy\,dx$$

$$I = \frac{\pi}{2} \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} 1\, dy\,dx$$

$$I = \frac{\pi}{2} \int_{x=0}^{a} |y|_0^{\sqrt{a^2-x^2}} dx$$

$$I = \frac{\pi}{2} \int_0^a \sqrt{a^2-x^2}\, dx \longrightarrow (1)$$

Let $x = a\sin\theta \Rightarrow \sin\theta = \frac{x}{a} \Rightarrow \theta = \sin^{-1}\left(\frac{x}{a}\right)$

$$\Rightarrow dx = a\cos\theta \, . \, d\theta$$

Upper limit $\Rightarrow x = a \Rightarrow \theta = \sin^{-1}\left(\frac{a}{a}\right) = \frac{\pi}{2}$

Lower limit $\Rightarrow x = 0 \Rightarrow \theta = \sin^{-1}(0) = 0$

$$\therefore \; I = \frac{\pi}{2}\int_{0}^{\pi/2} \left(\sqrt{a^2 - a^2\sin^2\theta}\right)(a\cos\theta)\,d\theta$$

$$I = \frac{\pi}{2}\, a \int_{0}^{\pi/2} a\sqrt{1-\sin^2\theta}\,.\,\cos\theta\,.\,d\theta$$

$$I = \frac{\pi}{2}\, a^2 \int_{0}^{\pi/2} \cos\theta\,.\,d\theta$$

$$I = \frac{\pi a^2}{2}\left|\frac{2-1}{2}\cdot\frac{\pi}{2}\right|$$

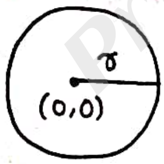
$$I = \frac{\pi a^2}{2}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

$$I = \frac{\pi^2 a^2}{8}$$

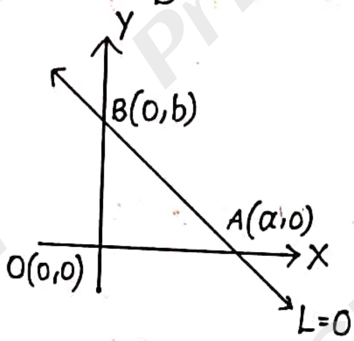
Change of Order of integration using double integration:

Some important curves and their shapes:

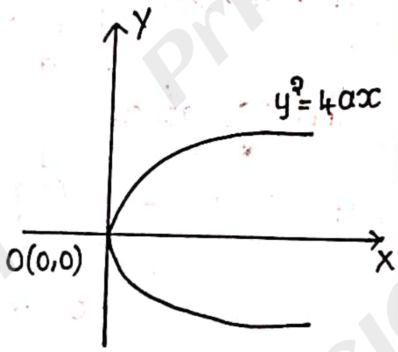
1) a) $x^2+y^2=r^2$



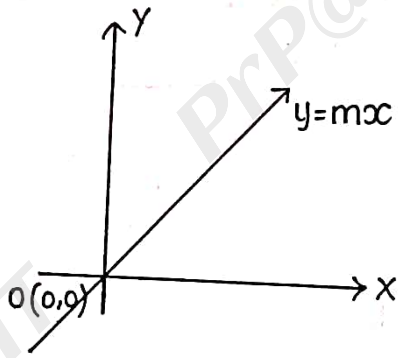
b) $\dfrac{x}{a}+\dfrac{y}{b}=1$



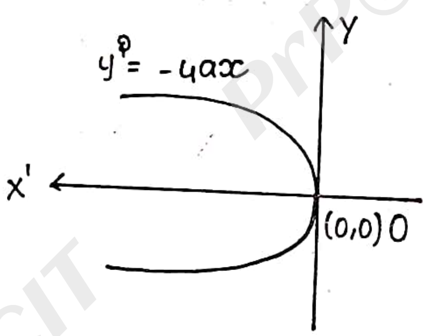
c) $y^2=4ax$



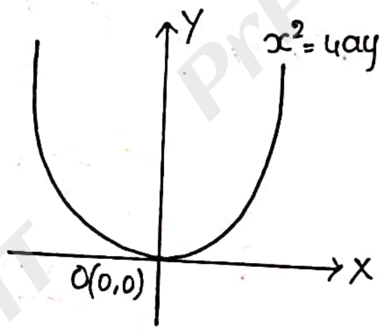
d) $y=mx$



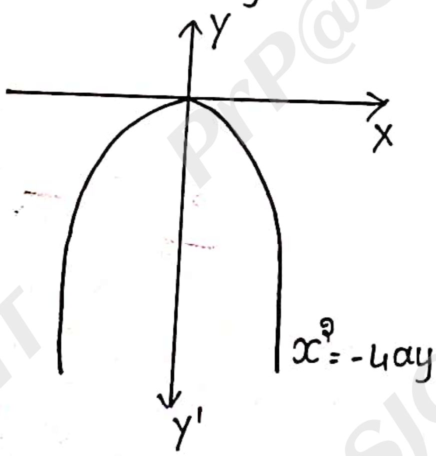
e) $y^2=-4ax$



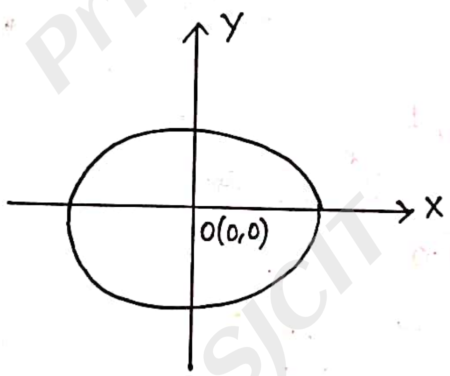
f) $x^2=4ay$



g) $x^2=-4ay$



h) $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$

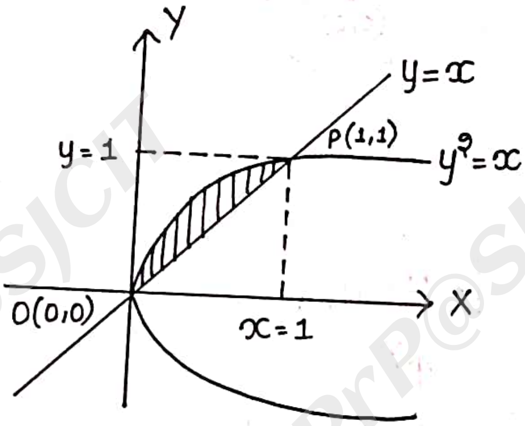


2) Let $I=\displaystyle\int_{x=a}^{b}\int_{y=f_1(x)}^{f_2(x)}\phi(x,y)\,dy\,dx$

plot all the boundaries $x=a$, $x=b$, $y=f_1(x)$, $y_2=f_2(x)$ on the graph and identify the common bounded region.

Rewrite the given integrals, $I = \int_{y=c}^{d} \int_{x=g_1(y)}^{g_2(y)} \phi(x,y)\, dx\, dy$ and simplify the same.

1. Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy\, dy\, dx$ by using change of order of integration.



Sol:- Let $I = \int_0^1 \int_x^{\sqrt{x}} xy\, dy\, dx \longrightarrow (1)$

where $x=0,\ x=1,\ y=x,\ y=\sqrt{x}$
$\Rightarrow y^2 = x$

$$\therefore (1) \Rightarrow I = \int_{y=0}^{1} \int_{x=y^2}^{y} xy\, dx\, dy$$

$$I = \int_{y=0}^{1} y \left|\frac{x^2}{2}\right|_{x=y^2}^{y} dy$$

$$I = \int_{y=0}^{1} y \left|\frac{y^2}{2} - \frac{y^4}{2}\right| dy$$

$$I = \frac{1}{2}\int_{y=0}^{1} (y^3 - y^5)\, dy$$

$$I = \frac{1}{2}\left|\frac{y^4}{4} - \frac{y^6}{6}\right|_0^1$$

$$I = \frac{1}{2}\left|\frac{1}{4} - \frac{1}{6}\right| \Rightarrow I = \frac{1}{2}\left|\frac{6-4}{24}\right|$$

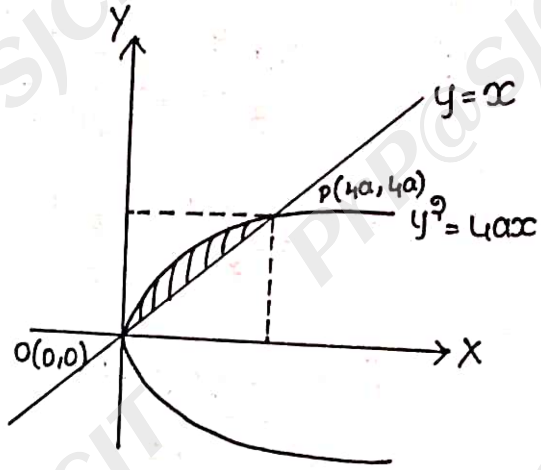
$$I = \frac{1}{\cancel{2}}\left|\frac{\cancel{2}}{24}\right| \Rightarrow I = \frac{1}{24}$$

2. Evaluate $\int_0^{4a}\int_x^{2\sqrt{ax}} x^2\,dy\,dx$ using change of order of integration.

Sol:- Let $I = \int_0^{4a}\int_x^{2\sqrt{ax}} x^2\,dy\,dx \longrightarrow (1)$

where $x=0,\ x=4a,\ y=x,\ y=2\sqrt{ax}$

$\Rightarrow y^2 = 4ax$



$\therefore (1) \Rightarrow I = \int_{y=0}^{4a}\int_{x=y}^{y^2/4a} x^2\,dx\,dy$

$$I = \int_{y=0}^{4a}\left|\frac{x^3}{3}\right|_y^{y^2/4a} dy$$

$$I = \frac{1}{3}\int_{y=0}^{4a}\left|x^3\right|_y^{y^2/4a} dy$$

$$I = \frac{1}{3}\int_{y=0}^{4a}\left|\left(\frac{y^2}{4a}\right)^3 - y^3\right| dy$$

$$I = \frac{1}{3}\int_{y=0}^{4a}\left|\frac{y^6}{64a^3} - y^3\right| dy$$

$$I = \frac{1}{3}\left\{\frac{1}{64a^3}\left|\frac{y^7}{7}\right|_0^{4a} - \left|\frac{y^4}{4}\right|_0^{4a}\right\}$$

$$I = \frac{1}{3}\left\{\frac{1}{64a^3}\cdot\frac{(4a)^7}{7} - \frac{(4a)^4}{4}\right\}$$

$$I = \frac{(4a)^4}{3}\left|\frac{(4a)^3}{64a^3\times 7} - \frac{1}{4}\right|$$

$$I = \frac{(4a)^4}{3}\left|\frac{1}{7} - \frac{1}{4}\right|$$

$$I = \frac{(4a)^4}{3}\left|\frac{4-7}{28}\right|$$

$$I = \frac{(4a)^4}{\cancel{3}}\left|\frac{-\cancel{3}}{28}\right|$$

$$I = \left|\frac{-256a^4}{28}\right|$$

$$I = \left|-\frac{64a^4}{7}\right|$$

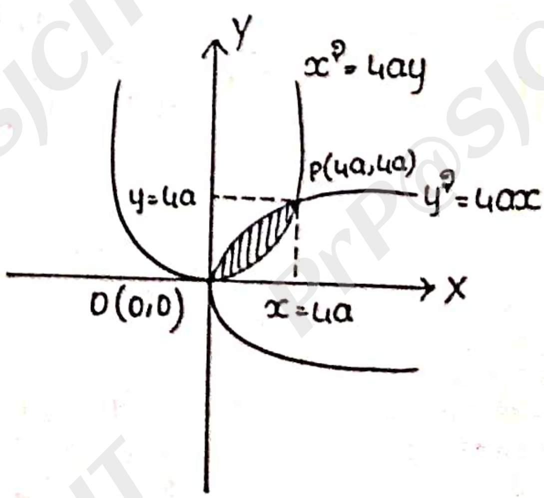
$$I = \frac{64a^4}{7}$$

3. Evaluate $\displaystyle\int_0^{4a}\int_{\frac{x^2}{4a}}^{2\sqrt{ax}} xy\, dy\, dx$ using change of order of integration.

Sol:- Let $\displaystyle I = \int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} xy\, dy\, dx \longrightarrow (1)$

where $x=0$, $x=4a$, $y=\dfrac{x^2}{4a} \Rightarrow x^2 = 4ay$, $y = 2\sqrt{ax}$
$\Rightarrow y^2 = 4ax$

and $x^2 = 4ay$, $y^2 = 4ax$ can intersect at $(0,0)$ & $(4a,4a)$



$$\therefore I = \int_{y=0}^{4a}\int_{x=y^2/4a}^{2\sqrt{ay}} xy\, dx\, dy$$

$$I = \int_{y=0}^{4a} y\left|\frac{x^2}{2}\right|_{y^2/4a}^{2\sqrt{ay}} dy$$

$$I = \frac{1}{2}\int_{y=0}^{4a} y\left|(2\sqrt{ay})^2 - \left(\frac{y^2}{4a}\right)^2\right| dy$$

$$I = \frac{1}{2}\int_{y=0}^{4a} y\left|4ay - \frac{y^4}{16a^2}\right| dy$$

$$I = \frac{1}{2}\int_{y=0}^{4a} \left|4ay^2 - \frac{y^5}{16a^2}\right| dy$$

$$I = \frac{1}{2}\left\{\frac{4ay^3}{3} - \frac{y^6}{16a^2\times 6}\right\}_0^{4a}$$

$$I = \frac{1}{2}\left\{\frac{4a(4a)^3}{3} - \frac{(4a)^6}{(4a)^2\times 6}\right\}$$

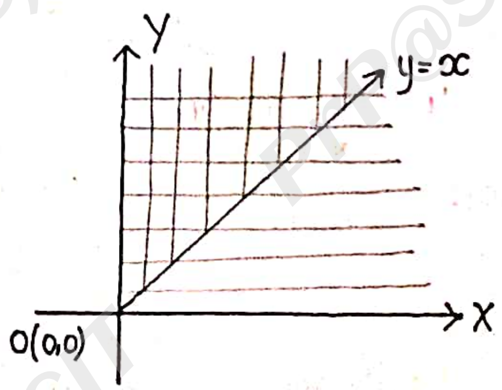
$$I = \frac{1}{2}\left|\frac{(4a)^4}{3} - \frac{(4a)^4}{6}\right|$$

$$I = \frac{(4a)^4}{2}\left|\frac{1}{3} - \frac{1}{6}\right|$$

$$I = \frac{256a^4}{2}\left|\frac{2-1}{6}\right|$$

$$I = \frac{64a^4}{3}$$

4. Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}\,dy\,dx$ using change of order of integration.



Sol:- Given, $I = \int_0^\infty \int_0^\infty \frac{e^{-y}}{y}\,dy\,dx \longrightarrow (1)$

where $x=0$, $x=\infty$, $y=x$, $y=\infty$

$\therefore (1) \Rightarrow I = \int_{y=0}^{\infty}\int_{x=0}^{y} \frac{e^{-y}}{y}\, dx\, dy$

$$I = \int_{y=0}^{\infty} \frac{e^{-y}}{y} \left| \int_{x=0}^{y} 1.dx \right| dy$$

$$I = \int_{y=0}^{\infty} \frac{e^{-y}}{y} \left| x \right|_{0}^{y} dy$$

$$\Rightarrow I = \int_{y=0}^{\infty} \frac{e^{-y}}{\cancel{y}} . \cancel{y}\, dy$$

$$I = \int_{y=0}^{\infty} e^{-y}. dy$$

$$I = \left| \frac{e^{-y}}{-1} \right|_{0}^{\infty}$$

$$I = -\left| e^{-\infty} - e^{0} \right|$$

$$I = -(0-1)$$

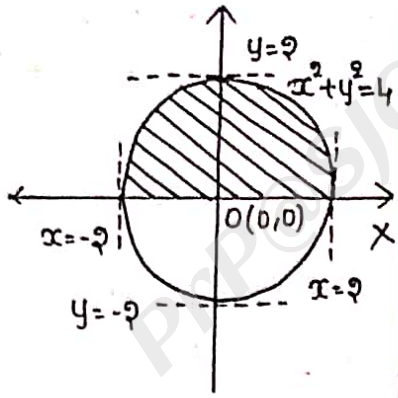
$$\boxed{I = 1}$$

5. Evaluate $\int_{-2}^{2}\int_{0}^{\sqrt{4-x^2}} (2-x)\, dy\, dx$ using change of order of integration.

Sol:- Given $I = \int_{-2}^{2}\int_{0}^{\sqrt{4-x^2}} (2-x)\, dy\, dx \longrightarrow (1)$

where $x=-2$, $x=2$ & $y=0$, $y=\sqrt{4-x^2} \Rightarrow 4-x^2=y^2 \Rightarrow x^2+y^2=2^2$

$$\therefore\ I = 2\int_{y=0}^{2}\int_{x=0}^{\sqrt{4-y^2}} (2-x)\,dx\,dy$$



$$I = 2\int_{y=0}^{2}\left|2x - \frac{x^2}{2}\right|_0^{\sqrt{4-y^2}} dy$$

$$I = 2\int_{y=0}^{2}\left|2\sqrt{4-y^2} - \frac{(4-y^2)}{2}\right| dy$$

$$I = 4\int_{y=0}^{2}\sqrt{4-y^2}\,dy - \int_{y=0}^{2}(4-y^2)\,dy$$

$$I = 4I_1 - I_2$$

$$\therefore\ I_1 = \int_0^{2}\sqrt{4-y^2}\,dy \longrightarrow (2)$$

But $y = 2\sin\theta \Rightarrow \theta = \sin^{-1}\left(y/2\right)$

$dy = 2\cos\theta \cdot d\theta$

and $\theta \Rightarrow \theta = \pi/2$

$$\therefore\ I_1 = \int_0^{\pi/2}\sqrt{4-4\sin^2\theta}\ \cdot 2\cos\theta \cdot d\theta$$

$$= \int_0^{\pi/2} 2\cos\theta \cdot 2\cos\theta \cdot d\theta$$

$$= 4\int_0^{\pi/2}\cos^2\theta \cdot d\theta$$

$$= 4\cdot\left|\frac{2-1}{2}\cdot\pi/2\right|$$

$$I_1 = 4 \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$I_1 = \pi$$

$$I_2 = \int_0^2 (4-y^2)\, dy$$

$$I_2 = \left| 4y - \frac{y^3}{3} \right|_0^2$$

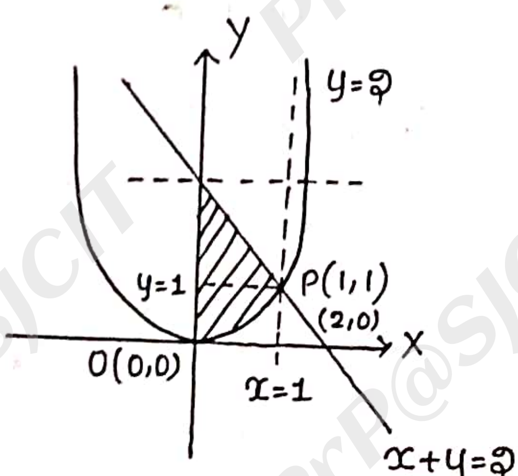
$$I_2 = 8 - \frac{8}{3}$$

$$I_2 = \frac{24-8}{3}$$

$$I_2 = \frac{16}{3}$$

$$\therefore \boxed{I = 4\pi - \frac{16}{3}}$$

6. Evaluate $\int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$ by changing the order of integration.



Sol:- $I = \int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$

where $x=0, x=1$ & $y=x^2$, $y=2-x$

$\Rightarrow x+y=2$

$\Rightarrow \frac{x}{2} + \frac{y}{2} = 1$

$$I = \int_{y=0}^{1} \int_{x=0}^{\sqrt{y}} xy\, dx\, dy + \int_{y=1}^{2} \int_{x=0}^{2-y} yx\, dy\, dx$$

$$I = I_1 + I_2$$

$$\therefore\ I_1 = \int_0^1 \int_0^{\sqrt{y}} xy\,dy\,dx \Rightarrow I_1 = \int_0^1 y\left[\frac{x^2}{2}\right]_0^{\sqrt{y}} dy$$

$$I_1 = \frac{1}{2}\int_{y=0}^{1} y(y-0)\,dy$$

$$I_1 = \frac{1}{2}\int_{y=0}^{1} y^2\,dy$$

$$I_1 = \frac{1}{2}\left|\frac{y^3}{3}\right|_0^1$$

$$I_1 = \frac{1}{2}\left(\frac{1}{3}\right)$$

$$\Rightarrow I_1 = \frac{1}{6}$$

$$\therefore\ I_2 = \int_{y=1}^{2}\int_{x=0}^{2-y} xy\,dx\,dy$$

$$I_2 = \int_{y=1}^{2} y\left[\frac{x^2}{2}\right]_0^{2-y} dy$$

$$I_2 = \frac{1}{2}\int_{y=1}^{2} y(y^2-4y+4)\,dy$$

$$I_2 = \frac{1}{2}\int_{y=1}^{2}(y^3-4y^2+4y)\,dy$$

$$I_2 = \frac{1}{2}\left[\frac{y^4}{4} - \frac{4y^3}{3} + \frac{4y^2}{2}\right]_1^2$$

$$I_2 = \frac{1}{2}\left\{\left[\frac{16}{4} - \frac{32}{3} + 8\right] - \left[\frac{1}{4} - \frac{4}{3} + 2\right]\right\}$$

$$I_2 = \frac{1}{2}\left[4 - \frac{32}{3} + 8 - \frac{1}{4} + \frac{4}{3} - 2\right]$$

$$I_2 = \frac{1}{2}\left[10 - \frac{28}{3} - \frac{1}{4}\right]$$

$$I_2 = \frac{1}{2}\left[\frac{120 - 112 - 3}{12}\right]$$

$$I_2 = \frac{1}{2} \cdot \frac{5}{12} = \frac{5}{24}$$

$$I = I_1 + I_2$$

$$I = \frac{1}{6} + \frac{5}{24}$$

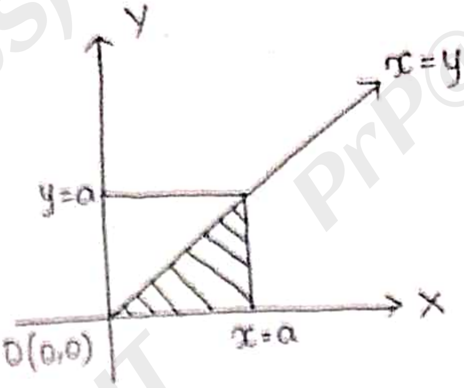
$$I = \frac{4+5}{24}$$

$$I = \frac{9}{24}$$

$$\therefore \boxed{I = \frac{3}{8}}$$

Evaluate $\int_0^a \int_y^a \frac{x}{x^2+y^2}\,dx\,dy$ by changing the order of integration.

Sol:- Let $I = \int_0^a \int_y^a \frac{x}{x^2+y^2}\,dx\,dy$



$$I = \int_{y=0}^{a} \int_{x=y}^{a} \frac{x}{x^2+y^2}\,dx\,dy$$

where $x=0$, $x=a$ & $y=0$, $y=a$

$\therefore$ By changing the order of integration.

$$I = \int_{x=0}^{a} \int_{y=0}^{x} \frac{x}{x^2+y^2}\,dy\,dx$$

$$I = \int_{x=0}^{a} x \int_{y=0}^{x} \frac{1}{x^2+y^2}\,dy\,dx$$

$$I = \int_{x=0}^{a} x \cdot \frac{1}{x} \left[\tan^{-1}\left(\frac{y}{x}\right)\right]_{y=0}^{x} dx$$

$$I = \int_{x=0}^{a} \left(\tan^{-1}\left(\frac{x}{x}\right) - \tan^{-1}(0)\right) dx$$

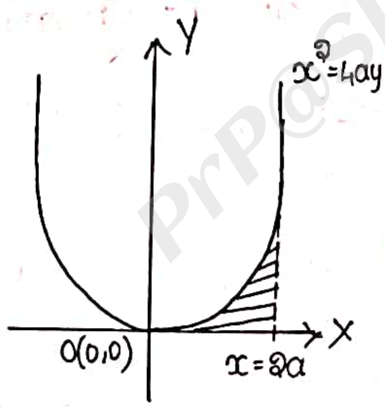
$$I = \int_{x=0}^{a} \frac{\pi}{4}\,dx$$

$$I = \frac{\pi}{4} \int_{x=0}^{a} dx$$

$$I = \frac{\pi}{4} |x|_0^a$$

$$\Rightarrow I = \frac{\pi a}{4}$$

Evaluate $\iint xy\,dx\,dy$ over the region bounded by the x-axis, ordinate $x = 2a$ and the curve $x^2 = 4ay$.



Sol:- Let $I = \int_{x=0}^{2a}\int_{y=0}^{x^2/4a} xy\,dx\,dy$

$$I = \int_{x=0}^{2a} x \int_{y=0}^{x^2/4a} y\,dy\,dx$$

$$I = \int_{x=0}^{2a} x \left[\frac{y^2}{2}\right]_0^{\frac{x^2}{4a}} dx$$

$$I = \frac{1}{2}\int_{x=0}^{2a} x \left[\frac{x^4}{16a^2}\right] dx$$

$$I = \frac{1}{32a^2}\int_{x=0}^{2a} x^5\,dx$$

$$I = \frac{1}{32a^2}\left[\frac{x^6}{6}\right]_0^{2a}$$

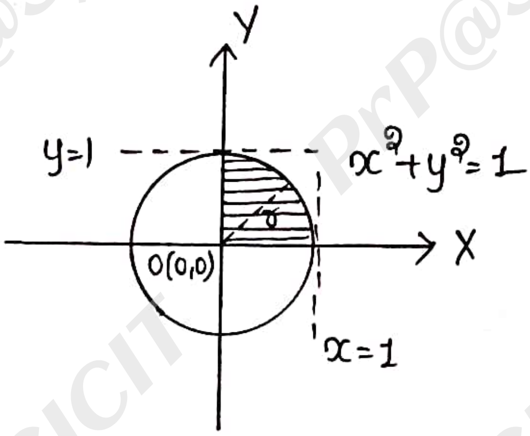
$$I = \frac{1}{32a^6}\left[\frac{(2a)^6}{6}\right]$$

$$I = \frac{1}{\cancel{32}a^{\cancel{2}}}\left[\frac{\cancel{64}^{\,2}a^{\cancel{6}}}{\cancel{6}_{\,3}}\right]$$

$$I = \frac{a^4}{3}$$

9. Evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2+y^2)\,dx\,dy$ by using polar coordinates.

Sol:- Let $I = \int_{y=0}^{1} \int_{x=0}^{\sqrt{1-y^2}} (x^2+y^2)\,dx\,dy$



where y varies from 0 to 1 &

x varies from $x=0$, $x=\sqrt{1-y^2}$

$\Rightarrow x^2 = 1-y^2$

$\Rightarrow x^2+y^2 = 1$

Let $x = r\cos\theta$, $y = r\sin\theta$

$\Rightarrow x^2+y^2 = r^2$ & $dx\,dy = r\,dr\,d\theta$

$\therefore I = \int_{r=0}^{1} \int_{\theta=0}^{\pi/2} r^2 . r . dr\,d\theta$

$\Rightarrow I = \int_{r=0}^{1} \int_{\theta=0}^{\pi/2} r^3\,dr\,d\theta$

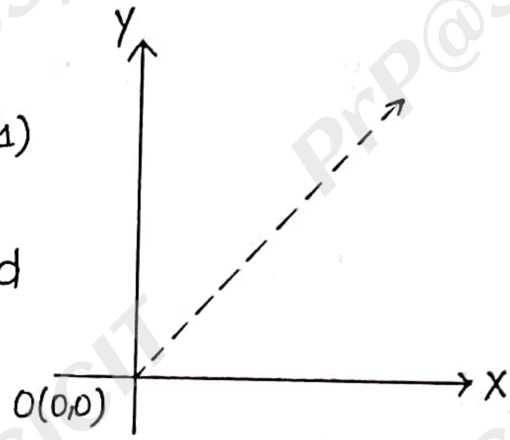
$\Rightarrow I = \int_{r=0}^{1} r^3 \left|\theta\right|_0^{\pi/2} dr$

$\Rightarrow I = \frac{\pi}{2} \int_{r=0}^{1} r^3 . dr$

$\Rightarrow I = \frac{\pi}{2} \left|\frac{r^4}{4}\right|_0^1$

$\Rightarrow I = \frac{\pi}{2} \times \frac{1}{4}$ $\Rightarrow \boxed{I = \frac{\pi}{8}}$

7. Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy\,dx$ by changing to polar coordinates.



Sol:- Let $I = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy\,dx \longrightarrow (1)$

where x varies from 0 to ∞ and y varies from 0 to ∞

But $x = r\cos\theta$, $y = r\sin\theta$

$\Rightarrow x^2 + y^2 = r^2$ and $dx\,dy = r\,dr\,d\theta$

Also 'r' varies from 0 to ∞ & 'θ' varies from 0 to $\pi/2$

$$\therefore I = \int_{r=0}^{\infty} \int_{\theta=0}^{\pi/2} e^{-r^2} . r\,dr\,d\theta .$$

$$I = \int_{r=0}^{\infty} e^{-r^2} . r . |\theta|_0^{\pi/2} dr$$

$$I = \frac{\pi}{2} \int_{r=0}^{\infty} e^{-r^2} . r . dr \longrightarrow (2)$$

but $r^2 = u$

$2r.dr = du$

$r.dr = \frac{1}{2} du$

upper limit $\Rightarrow r = \infty \Rightarrow u = \infty$

lower limit $\Rightarrow r = 0 \Rightarrow u = 0$

$$I = \frac{\pi}{2}\int_{u=0}^{\infty} e^{-u}\cdot\frac{1}{2}du$$

$$I = \frac{\pi}{2}\int_{u=0}^{\infty} e^{-u}\cdot\frac{1}{2}\,du$$

$$I = \frac{\pi}{4}\left[\frac{e^{-u}}{-1}\right]_{u=0}^{\infty}$$

$$I = \frac{-\pi}{4}\left[e^{-\infty}-e^{0}\right]$$

$$I = \frac{-\pi}{4}(0-1)$$

$$I = \frac{\pi}{4}$$

$$\Rightarrow \int_{x=0}^{\infty}\int_{y=0}^{\infty} e^{-(x^2+y^2)}\,dy\,dx = \frac{\pi}{4}$$

$$\int_{x=0}^{\infty} e^{-x^2}\cdot dx \int_{y=0}^{\infty} e^{-y^2}\cdot dy = \frac{\pi}{4} \longrightarrow (3)$$

But $y = x$

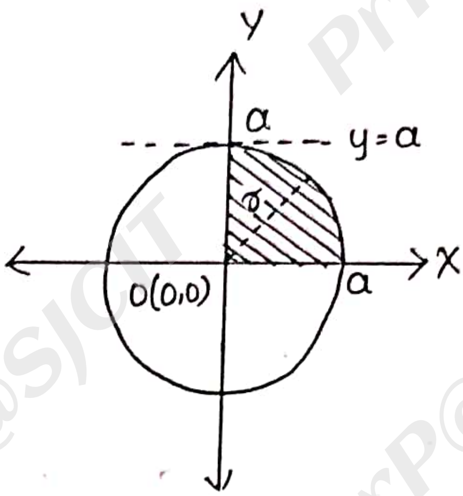
$$\therefore (3) \Rightarrow \int_{x=0}^{\infty} e^{-x^2}\cdot dx \int_{x=0}^{\infty} e^{-x^2} dx = \frac{\pi}{4}$$

$$\Rightarrow \left[\int_0^{\infty} e^{-x^2}\cdot dx\right]^2 = \frac{\pi}{4}$$

$$\Longrightarrow \int_0^{\infty} e^{-x^2}\cdot dx = \frac{\sqrt{\pi}}{2}$$

8. Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} y(\sqrt{x^2+y^2})\,dx\,dy$ by changing to polar co-ordinatives.

Sol:- Let $I = \int_0^a \int_0^{\sqrt{a^2-y^2}} y\sqrt{x^2+y^2}\,dx\,dy$



where y varies from 0 to a &

x varies from 0 to $\sqrt{a^2-y^2}$

$$x = \sqrt{a^2-y^2}$$

$$x^2 = a^2-y^2$$

$$\Rightarrow x^2+y^2 = a^2$$

(Equation of Circle)

$\therefore$ Let $x = r\sin\theta$, $y = r\sin\theta$

$$x^2+y^2 = r^2$$

$$dx\,dy = r\,dr\,d\theta$$

And r varies from 0 to a & θ varies from 0 to $\frac{\pi}{2}$

$$I = \int_{r=0}^{a} \int_{\theta=0}^{\pi/2} (r\sin\theta)\, r.r\,dr.d\theta$$

$$I = \int_{r=0}^{a} \int_{\theta=0}^{\pi/2} r^3 \sin\theta\, dr\,d\theta$$

$$I = \int_{r=0}^{a} r^3 \left| \int_0^{\pi/2} \sin\theta . d\theta \right| dr$$

$$I = \int_{r=0}^{a} r^3 \left[-\cos\theta\right]_0^{\pi/2} dr$$

$$I = \int_{r=0}^{a} r^3 \left[-\cos\frac{\pi}{2} + \cos 0\right] dr$$

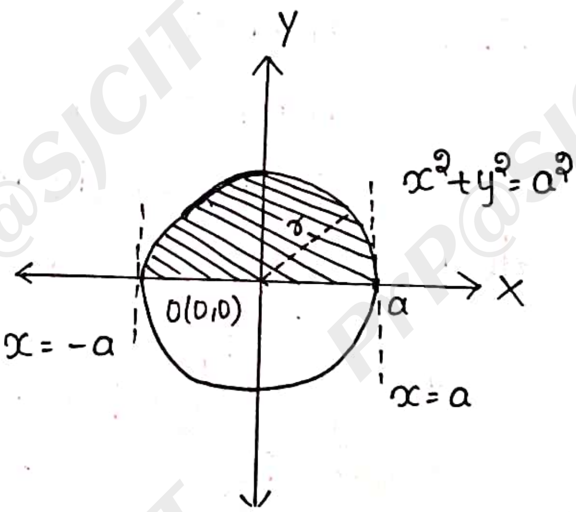
$$I = \int_{r=0}^{a} r^3 . dr$$

$$I = \left[\frac{r^4}{4}\right]_0^a$$

$$\boxed{I = \frac{a^4}{4}}$$

9. Evaluate $\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2}\, dx\, dy$.

Sol:- $I = \int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2}\, dx\, dy$



where x varies from $-a$ to a

y varies from 0 to $\sqrt{a^2-x^2}$

$$y = \sqrt{a^2-x^2}$$

$$y^2 = a^2 - x^2$$

$$x^2+y^2 = a^2$$

(represents a circle)

$\therefore$ Let $x = r\cos\theta$, $y = r\sin\theta$

$$x^2+y^2 = r^2$$

$$dy\,dx = r.dr\,d\theta$$

And r varies from 0 to a,

θ varies from 0 to π

$$I = \int_{r=0}^{a}\int_{\theta=0}^{\pi} r \cdot r \, dr \, d\theta$$

$$I = \int_{r=0}^{a}\int_{\theta=0}^{\pi} r^2 \, dr \, d\theta$$

$$I = \int_{r=0}^{a} r^2 \left[\theta\right]_0^{\pi} dr$$

$$I = \pi \int_{r=0}^{a} r^2 \cdot dr$$

$$I = \pi \left[\frac{r^3}{3}\right]_0^{a}$$

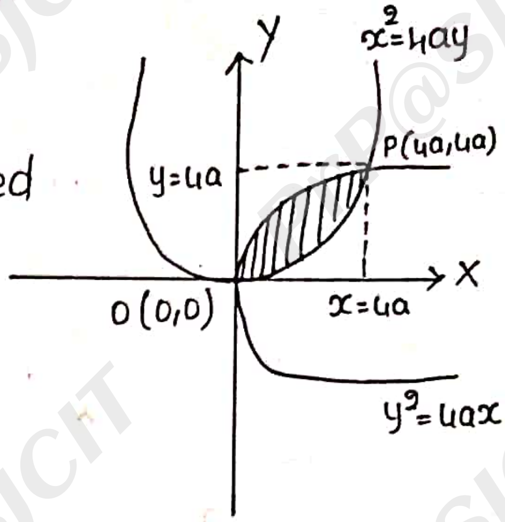
$$\boxed{I = \frac{\pi a^3}{3}}$$

Applications:-

The main aim of applications are to find the area of the surface using double integration and the volume of tetrahedron by using triple or double integral.

They are $A = \iint dx\,dy$ and $V = \iiint dx\,dy\,dz$

1. Find the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ using double integration.



Sol:- Given, $y^2 = 4ax$ & $x^2 = 4ay$ be the parabolas then the area bounded between $y^2 = 4ax$ & $x^2 = 4ay$.

$$A = \iint dx\,dy$$

$$\Rightarrow A = \int_{x=0}^{4a} \int_{y=\frac{x^2}{4a}}^{2\sqrt{ax}} dy\,dx$$

$$A = \int_{x=0}^{4a} \left| y \right|_{\frac{x^2}{4a}}^{2\sqrt{ax}} dx$$

$$A = \int_{x=0}^{4a} \left| 2\sqrt{ax} - \frac{x^2}{4a} \right| dx$$

$$A = 2\sqrt{a} \int_{x=0}^{4a} x^{1/2}.\,dx - \frac{1}{4a} \int_{x=0}^{4a} x^2.\,dx$$

$$A = 2\sqrt{a} \left| \frac{x^{3/2}}{3/2} \right|_0^{4a} - \frac{1}{4a} \left| \frac{x^3}{3} \right|_0^{4a}$$

$$A = \frac{4\sqrt{a}}{3} \left[x^{3/2} \right]_0^{4a} - \frac{1}{12a} \left[x^3 \right]_0^{4a}$$

$$A = \frac{4\sqrt{a}}{3} (4a)^{3/2} - \frac{1}{12a} (4a)^3$$

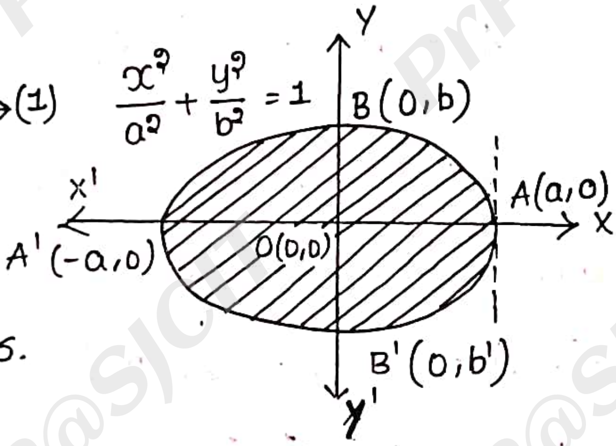
$$A = \frac{(4\sqrt{a})(4a)(4a)^{1/2}}{3} - \frac{(4a)(4a)^2}{3(4a)}$$

$$A = \frac{(4\sqrt{a})(4a)(2\sqrt{a})}{3} - \frac{16a^2}{3}$$

$$A = \frac{32a^2}{3} - \frac{16a^2}{3}$$

$$\Rightarrow A = \frac{16a^2}{3} \text{ Sq. units}$$

2. Find the area of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using double integration.



Sol:- Given, Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow (1)$

has the vertex as origin and the co-ordinate axis are dividing into four equal parts.

$\therefore$ area of ellipse is $A = 4A$

$$A = 4\int_{x=0}^{a}\int_{y=0}^{b/a\sqrt{a^2-x^2}} dy\,dx$$

$$A = 4\int_{x=0}^{a} [y]_{0}^{b/a\sqrt{a^2-x^2}} dx$$

$$A = \frac{4b}{a}\int_{x=0}^{a} \sqrt{a^2-x^2}\, dx \rightarrow (2)$$

Let $x = a\sin\theta \Rightarrow \sin\theta = \frac{x}{a} \Rightarrow \theta = \sin^{-1}\left(\frac{x}{a}\right)$

$\therefore dx = a\cos\theta . d\theta$

upper limit : $x = a \Rightarrow \theta = \pi/2$

lower limit : $x = 0 \Rightarrow \theta = 0$

$$\therefore A = \frac{4b}{a}\int_0^{\pi/2}\sqrt{a^2 - a^2\sin^2\theta}\,.\,a\cos\theta.d\theta$$

$$A = \frac{4b}{a}\int_0^{\pi/2} a\cos\theta.a\cos\theta.\,d\theta$$

$$A = 4ab\int_0^{\pi/2}\cos^2\theta.d\theta$$

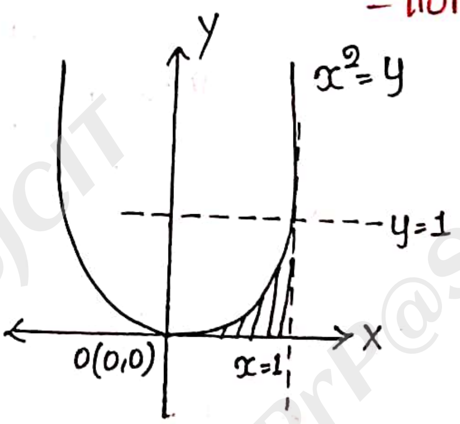
$$A = 4ab.\left|\frac{2-1}{2}.\frac{\pi}{2}\right|$$

$$\Rightarrow A = \pi ab \text{ sq.units.}$$

where $b = a$, then eq-(1) represents a circle $x^2 + y^2 = a^2$

$\therefore A = \pi.a.a \Rightarrow \pi a^2$ sq.units.

3. Evaluate $\int_0^1\int_{\sqrt{y}}^1 dx\,dy$ by changing the order of integra-tion.



Sol:- $$I = \int_{y=0}^{y=1}\int_{x=\sqrt{y}}^{1} dx\,dy$$

where $y = 0$ to $y = 1$ &

$x = \sqrt{y} \Rightarrow x^2 = y$, $x = 1$

$$\therefore I = \int_{x=0}^{1}\int_{y=0}^{x^2} 1.dy\,dx$$

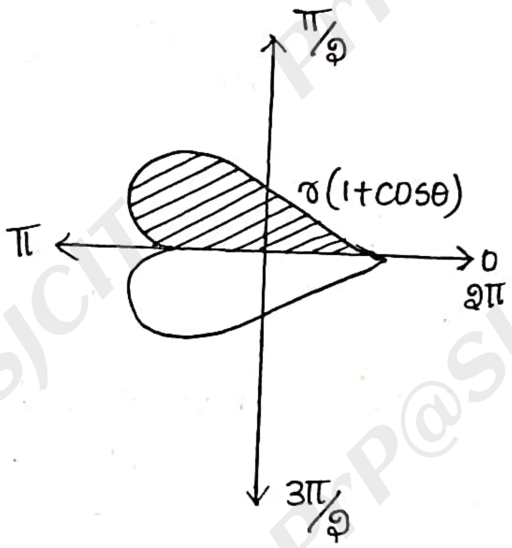
$$I = \int_{x=0}^{1} |y|_0^{x^2} dx$$

$$I = \int_{x=0}^{1} x^2 dx$$

$$\Rightarrow I = \left|\frac{x^3}{3}\right|_0^1$$

$$\boxed{I = \frac{1}{3}}$$

4. Find the area of Cardiod $r = a(1+\cos\theta)$ about the initial line.



Sol:- $A = \iint r.dr.d\theta$

$$A = \int_{r=0}^{a(1+\cos\theta)} \int_{\theta=0}^{\pi} r.dr.d\theta$$

$$A = \int_{\theta=0}^{\pi} \int_{r=0}^{a(1+\cos\theta)} r.dr.d\theta$$

$$A = \int_{\theta=0}^{\pi} \left|\frac{r^2}{2}\right|_0^{a(1+\cos\theta)} d\theta$$

$$A = \frac{1}{2}\int_{\theta=0}^{\pi} a^2 (1+\cos\theta)^2 d\theta$$

$$A = \frac{a^2}{2}\int_{\theta=0}^{\pi} (2\cos^2\theta/2)^2 d\theta$$

$$A = \frac{4a^2}{2}\int_{\theta=0}^{\pi} \cos^4(\theta/2)\, d\theta \longrightarrow (1)$$

Let $\theta/2 = u$

$\Rightarrow \theta = 2u$

$\Rightarrow d\theta = 2du.$

upper limit: $\theta = \pi \Rightarrow u = \pi/2$

lower limit: $\theta = 0 \Rightarrow u = 0$

$$\therefore A = 2a^2 \int_0^{\pi/2} \cos^4 u . 2u$$

$$A = 4a^2 \int_0^{\pi/2} \cos^4 u . du$$

$$A = 4a^2 . \frac{4-1}{4} . \frac{4-3}{4-2} . \frac{\pi}{2}$$

$$\Rightarrow A = \cancel{4}a^2 . \frac{3}{\cancel{4}} . \frac{1}{2} . \frac{\pi}{2}$$

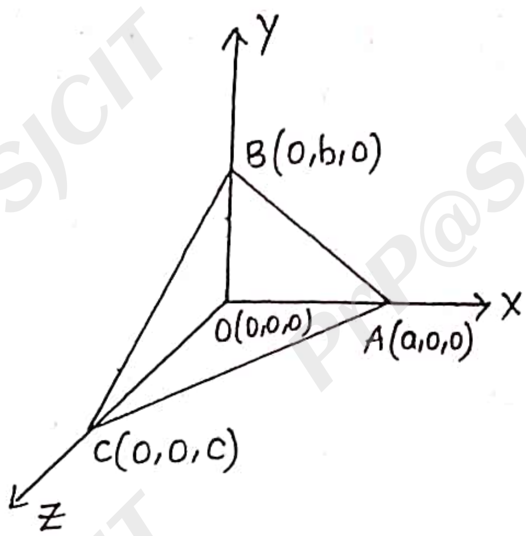
$$\boxed{A = \frac{3\pi a^2}{4} \text{ sq.units}}$$

5. Find the volume of the tetrahedron $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ bounded by the planes $x=0,\ y=0,\ z=0.$

Sol:- $\therefore V = \iiint dx\,dy\,dz$

$$V = \int_{x=0}^{a} \int_{y=0}^{\frac{b}{a}(a-x)} \int_{z=0}^{c\left[1-\frac{x}{a}-\frac{y}{b}\right]} 1 . dz\,dy\,dx$$

$$V = \int_{x=0}^{a}\int_{y=0}^{\frac{b}{a}(a-x)} |z|_{0}^{c\left[1-\frac{x}{a}-\frac{y}{b}\right]} \, dy\,dx$$



$$V = c\int_{x=0}^{a}\int_{y=0}^{\frac{b}{a}(a-x)} \left[1-\frac{x}{a}-\frac{y}{b}\right] dy\,dx$$

$$V = c\int_{x=0}^{a}\left[y-\frac{xy}{a}-\frac{y^2}{b}\right]_{0}^{\frac{b}{a}(a-x)} dx$$

$$V = c\int_{x=0}^{a}\left|\frac{b}{a}(a-x)-\frac{x}{a}\cdot\frac{b}{a}(a-x)-\frac{1}{2\cancel{b}}\cdot\frac{b^{\cancel{2}}}{a^2}(a-x)^2\right| dx$$

$$V = c\int_{x=0}^{a}\left\{\frac{b}{a}(a-x)\left[1-\frac{x}{a}\right]-\frac{b}{2a^2}(a-x)^2\right\} dx$$

$$V = c\int_{x=0}^{a}\left|\frac{b}{a}(a-x)\frac{(a-x)}{a}-\frac{b}{2a^2}(a-x)^2\right\} dx$$

$$V = c\int_{x=0}^{a}\frac{b}{a^2}(a-x)^2-\frac{b}{2a^2}(a-x)^2$$

$$V = \frac{c}{2}\int_{x=0}^{a}\frac{b}{a^2}(a-x)^2\cdot dx$$

$$V = \frac{c}{2}\int_{x=0}^{a}\frac{b}{a^2}(x-a)^2\,dx$$

$$V = \frac{bc}{2a^2}\int_{x=0}^{a}(x-a)^2\,dx$$

$$V = \frac{bc}{2a^2}\left[\frac{(x-a)^3}{3}\right]_0^a$$

$$V = \frac{bc}{2a^2}\left|0-\frac{(-a)^3}{3}\right|$$

$$\Rightarrow V = \frac{bc}{2\cancel{a^2}} \times \frac{a^{\cancel{3}}}{3} \quad \Rightarrow V = \frac{abc}{6}$$

$$\boxed{V = \frac{abc}{6} \text{ Cubic units}}$$

6. Find the volume of the solid bounded by the plane $x=0$, $y=0$, $z=0$, $x+y+z=1$.

Sol:- $\therefore V = \iiint dx\,dy\,dz$

$$V = \int_{x=0}^{1}\int_{y=0}^{(1-x)}\int_{z=0}^{(1-x-y)} 1.\,dz\,dy\,dx$$

$$V = \int_{x=0}^{1}\int_{y=0}^{(1-x)} \left|z\right|_0^{(1-x-y)} dy\,dx$$

$$V = \int_{x=0}^{1}\int_{y=0}^{(1-x)} (1-x-y)\,dy\,dx$$

$$V = \int_{x=0}^{1} \left|1-xy-\frac{y^2}{2}\right|_0^{(1-x)} dx$$

$$V = \int_{x=0}^{1} \left|(1-x)-x(1-x)-\frac{1}{2}(1-x)^2\right| dx$$

$$V = \int_{x=0}^{1} \left[(1-x)(1-x) - \frac{1}{2}(1-x)^2 \right] dx$$

$$V = \int_{x=0}^{1} \left[(1-x)^2 - \frac{1}{2}(1-x)^2 \right] dx$$

$$V = \frac{1}{2} \int_{x=0}^{1} (1-x)^2 dx$$

$$V = \frac{1}{2} \int_{x=0}^{1} (x-1)^2 dx$$

$$V = \frac{1}{2} \left[\frac{(x-1)^3}{3} \right]_0^1$$

$$\Rightarrow V = \frac{1}{2} \left[\frac{0-(-1)^3}{3} \right] \Rightarrow V = \frac{1}{2} \cdot \frac{1}{3}$$

$$\boxed{V = \frac{1}{6}}$$

7. A pyramid is bounded by three co-ordinate axis planes and the plane $x + 2y + 3z = 6$. Compute the volume of plane.

Sol:- WKT,

$$V = \iiint dz\,dy\,dx$$

$$V = \int_{x=0}^{6} \int_{y=0}^{\frac{1}{2}(6-x)} \int_{z=0}^{\frac{1}{3}(6-x-2y)} 1 \cdot dz\,dy\,dx$$

$$V = \int_{x=0}^{6} \int_{y=0}^{\frac{1}{2}(6-x)} |z|_0^{\frac{1}{3}(6-x-2y)} dy\,dx$$

$$\Rightarrow V = \frac{1}{3}\int_{x=0}^{6}\int_{y=0}^{\frac{1}{2}(6-x)} (6 - x - 2y)\, dy\, dx$$

$$\Rightarrow V = \frac{1}{3}\int_{x=0}^{6} \left| 6y - xy - y^2 \right|_{y=0}^{\frac{1}{2}(6-x)} dx$$

$$\Rightarrow V = \frac{1}{3}\int_{x=0}^{6} \left| 6\cdot\frac{1}{2}(6-x) - x\cdot\frac{1}{2}(6-x) - \frac{1}{4}(6-x)^2 \right| dx$$

$$\Rightarrow V = \frac{1}{3}\int_{x=0}^{6} \left[\frac{1}{2}(6-x)(6-x) - \frac{1}{4}(6-x)^2 \right] dx$$

$$\Rightarrow V = \frac{1}{3}\int_{x=0}^{6} \left| \frac{1}{2}(6-x)^2 - \frac{1}{4}(6-x)^2 \right| dx$$

$$\Rightarrow V = \frac{1}{3}\int_{x=0}^{6} \left(\frac{1}{2} - \frac{1}{4}\right)(6-x)^2 dx$$

$$\Rightarrow V = \frac{1}{3\times 4}\int_{0}^{6} (x-6)^2 dx$$

$$\Rightarrow V = \frac{1}{12}\left| \frac{(x-6)^3}{3} \right|_{0}^{6}$$

$$\Rightarrow V = \frac{1}{12}\left| 0 - \frac{(-6)^3}{3} \right|$$

$$\Rightarrow V = \frac{1}{2\times 6} \times \frac{6^3}{3}$$

$\Rightarrow V = 6$ Cubic units.

8. Find the volume bounded by the cylinder $x^2+y^2=4$ and the planes $y+z=4$ and $z=0$.

Sol:- Given the cylinder $x^2+y^2=4$ is placed in XY-plane bounded by the planes $y+z=4$, $z=0$.

$\therefore z=4-y$

x varies from 0 to $\sqrt{4-y^2}$ and

y varies from -2 to $+2$

$\therefore$ We have the volume using double integration

$$V = 2\iint z\,dx\,dy$$

$$V = 2\int_{y=-2}^{2}\int_{x=0}^{\sqrt{4-y^2}} (4-y)\,dx\,dy$$

$$V = 2\int_{y=-2}^{2}(4-y)\int_{x=0}^{\sqrt{4-y^2}} 1.\,dx\,dy$$

$$V = 2\int_{y=-2}^{2}(4-y)\,[x]_0^{\sqrt{4-y^2}}\,dy$$

$$V = 2\int_{y=-2}^{2}(4-y)\sqrt{4-y^2}\,dy$$

$$V = 2\int_{y=-2}^{2}4\sqrt{4-y^2}\,dy - \int_{y=-2}^{2}y\sqrt{4-y^2}\,dy$$

$$V = 8\int_{y=-2}^{2}\sqrt{4-y^2}\,dy - 0 \qquad (\because y\sqrt{4-y^2} \text{ is odd})$$

$$V = 8\int_{-2}^{2}\sqrt{4-y^2}\,dy$$

$$V = 8\times 2\int_{0}^{2}\sqrt{4-y^2}\,dy \quad (\because \sqrt{4-y^2} \text{ is even})$$

$$V = 8\times 2\int_{0}^{2}\sqrt{4-y^2}\,dy$$

$$V = 16\int_{0}^{2}\sqrt{4-y^2}\,dy$$

Let $y = 2\sin\theta \Rightarrow \theta = \sin^{-1}\left(\frac{y}{2}\right)$

$dy = 2\cos\theta \cdot d\theta$

upper limit : $y = 2 \Rightarrow \theta = \frac{\pi}{2}$

lower limit : $y = 0 \Rightarrow \theta = 0$

$$V = 16\int_{0}^{\pi/2}\sqrt{4-4\sin^2\theta}\cdot 2\cos\theta\,d\theta$$

$$V = 16\int_{0}^{\pi/2} 2\sqrt{1-\sin^2\theta}\cdot 2\cos\theta\,d\theta$$

$$V = 32\int_{0}^{\pi/2} 2\cos^2\theta\,d\theta$$

$$V = \cancel{2}\times\overset{16}{\cancel{32}}\cdot\left(\frac{2-1}{\cancel{2}}\right)\cdot\frac{\pi}{\cancel{2}}$$

$\Rightarrow V = 16\pi$ Cubic units.

Beta - Gamma function:

Let m, n be the two positive number then an improper integral $\int_0^1 x^{m-1}\cdot(1-x)^{n-1}\,dx$ is called the beta function and which can be defined as

$$\beta(m,n) = \int_0^1 x^{m-1}\cdot(1-x)^{n-1}\,dx \longrightarrow (1)$$

To get the simplified form,

Let $x = \sin^2\theta \Rightarrow \sin\theta = \sqrt{x} \Rightarrow \theta = \sin^{-1}(\sqrt{x})$

$\Rightarrow dx = 2\sin\theta.\cos\theta\, d\theta$

$\therefore$ UL: $x = 1 \Rightarrow \theta = \pi/2$

LL: $x = 0 \Rightarrow \theta = 0$

$$\therefore (1) \Rightarrow \beta(m,n) = \int_0^{\pi/2} (\sin^2\theta)^{m-1}\cdot(1-\sin^2\theta)^{n-1}\cdot 2\sin\theta.\cos\theta\,.d\theta$$

$$= 2\int_0^{\pi/2} (\sin^2\theta)^{m-1}(\cos^2\theta)^{n-1}\cdot\sin\theta.\cos\theta.d\theta$$

$$= 2\int^{\pi/2} \sin^{2m-1}\cdot\cos^{2n-1}\,d\theta$$

To get more simplification,

Let $P = 2m-1 \Rightarrow m = \frac{P+1}{2}$

$q = 2n-1 \Rightarrow n = \frac{q+1}{2}$

$$\therefore (2) \Rightarrow \beta\left(\frac{P+1}{2}, \frac{q+1}{2}\right) = 2\int_0^{\pi/2} \sin^P\theta\,.\cos^q\theta.d\theta$$

$$\Rightarrow \int_0^{\pi/2} \sin^P\theta.\cos^q\theta.d\theta = \frac{1}{2}\beta\left(\frac{P+1}{2}, \frac{q+1}{2}\right)$$

2. For any $n>0$, an improper integral $\int_0^{\infty} e^{-x} \cdot x^{n-1} \cdot dx$ is called the Gamma function & it is defined as

$$\Gamma n = \int_0^{\infty} e^{-x} \cdot x^{n-1} \cdot dx$$

To get the simplified form

Let $x = y^2 \Rightarrow y = \sqrt{x}$

$\Rightarrow dx = 2y\,dy$

UL : $x = \infty \Rightarrow y = \infty$

LL : $x = 0 \rightarrow y = 0$

$$\therefore (1) \Rightarrow \Gamma n = \int_0^{\infty} e^{-y^2} \cdot (y^2)^{n-1} \cdot 2y\,dy.$$

$$\Rightarrow \Gamma n = 2\int_0^{\infty} e^{-y^2} \cdot y^{2n-1} dy$$

$$\Rightarrow \Gamma n = 2\int_0^{\infty} e^{-x^2} \cdot x^{2n-1} dx \quad (\because y = x)$$

3. Relation between Beta and Gamma function:-

WKT, $\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cdot \cos^{2n-1}\theta \cdot d\theta =$

$$2\int_0^{\pi/2} \sin^{2n-1}\theta \cdot \cos^{2m-1}\theta \cdot d\theta \longrightarrow (1)$$

$$\Gamma m = 2\int_0^{\infty} e^{-x^2} \cdot x^{2m-1} dx \longrightarrow (2)$$

$$\Gamma n = 2\int_0^{\infty} e^{-y^2} \cdot y^{2n-1} dy \longrightarrow (3)$$

$$\Gamma(m+n) = 2\int_0^{\infty} e^{-r^2} \cdot r^{2m-1} \cdot dr \longrightarrow (4)$$

Let

$$\Gamma m \cdot \Gamma n = 2\int_{x=0}^{\infty} e^{-x^2} \cdot x^{2m-1} \cdot dx \cdot 2\int_{y=0}^{\infty} e^{-y^2} \cdot y^{2n-1} \cdot dy$$

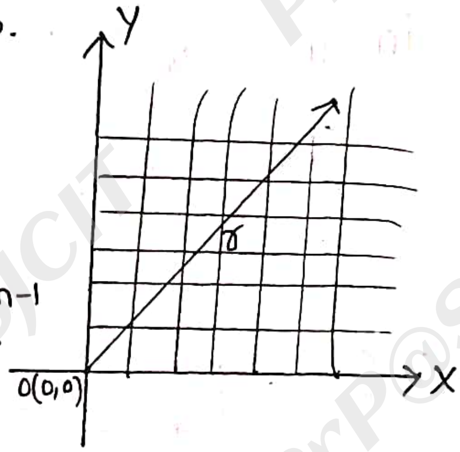
$$\Gamma m \cdot \Gamma n = 4\int_{x=0}^{\infty}\int_{y=0}^{\infty} e^{-x^2} \cdot e^{-y^2} \cdot x^{2m-1} \cdot y^{2n-1}\, dx\,dy$$

$$\Gamma m \cdot \Gamma n = 4\int_{x=0}^{\infty}\int_{y=0}^{\infty} e^{-(x^2+y^2)} \cdot x^{2m-1} \cdot y^{2n-1}\, dx\,dy \longrightarrow (5)$$

By changing to polar coordinates.

Let $x = r\cos\theta$, $y = r\sin\theta$

$\Rightarrow x^2+y^2 = r^2$ & $dx\,dy = r\,dr\,d\theta$



$$\therefore (5) \Rightarrow \Gamma m\,\Gamma n = 4\int_{r=0}^{\infty}\int_{\theta=0}^{\pi/2} e^{-r^2} \cdot (r\cos\theta)^{2m-1}(r\sin\theta)^{2n-1} \cdot r\,dr\,d\theta.$$

$$\Rightarrow \Gamma m\,\Gamma n = 4\int_{r=0}^{\infty}\int_{\theta=0}^{\pi/2} e^{-r^2} \cdot r^{2(m+n)-1} \cdot \sin^{2n-1}\theta \cdot \cos^{2m-1}\theta\, dr\,d\theta$$

$$\Rightarrow \Gamma m \cdot \Gamma n = \left|2\int_0^{\infty} e^{-r^2} \cdot r^{2(m+n)-1} dr\right| \left|2\int_0^{\pi/2} \sin^{2n-1}\theta \cdot \cos^{2m-1}\theta \cdot d\theta\right|$$

$$\Rightarrow \Gamma m \cdot \Gamma n = \Gamma(m+n)\, \beta(m,n)$$

$$\Rightarrow \beta(m,n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma(m+n)}$$

4. Show that $\Gamma\frac{1}{2} = \sqrt{\pi}$.

WKT, $\Gamma m = 2\int_0^{\alpha} e^{-x^2} \cdot x^{2m-1} \cdot dx \longrightarrow (1)$

$$\Gamma n = 2\int_0^{\alpha} e^{-y^2} \cdot y^{2n-1} \cdot dy \longrightarrow (2)$$

Let

$$\Gamma m \cdot \Gamma n = 2\int_0^{\alpha} e^{-x^2} \cdot x^{2m-1} \cdot dx \cdot 2\int_0^{\alpha} e^{-y^2} \cdot y^{2n-1} \cdot dy$$

$$\Gamma m \cdot \Gamma n = 4\int_{x=0}^{\alpha}\int_{y=0}^{\alpha} e^{-x^2} \cdot e^{-y^2} \cdot x^{2m-1} \cdot y^{2n-1} \cdot dx\,dy$$

$$\Gamma m \cdot \Gamma n = 4\int_{x=0}^{\alpha}\int_{y=0}^{\alpha} e^{-(x^2+y^2)} \cdot x^{2m-1} \cdot y^{2n-1}\, dx\,dy \longrightarrow (3)$$

Let $m = \frac{1}{2}$ & $n = \frac{1}{2}$

$$\therefore (3) \Rightarrow \Gamma\tfrac{1}{2} \cdot \Gamma\tfrac{1}{2} = 4\int_{x=0}^{\alpha}\int_{y=0}^{\alpha} e^{-(x^2+y^2)} \cdot x^0 \cdot y^0\, dx\,dy$$

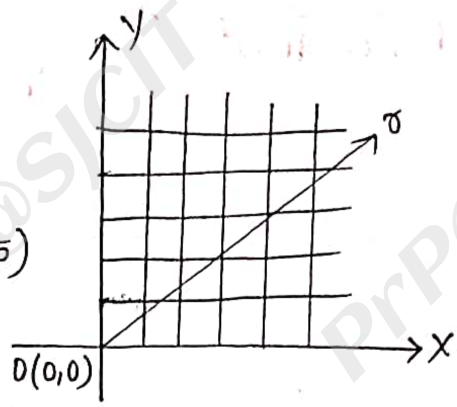
$$\Rightarrow \left(\Gamma\tfrac{1}{2}\right)^2 = 4\int_{x=0}^{\alpha}\int_{y=0}^{\alpha} e^{-(x^2+y^2)}\, dx\,dy$$

Let $x = r\cos\theta$, $y = r\sin\theta$

$$\Rightarrow x^2+y^2 = r^2 \text{ \& } dx\,dy = r\,dr\,d\theta$$

$$\therefore (4) \Rightarrow \left(\Gamma\tfrac{1}{2}\right)^2 = 4\int_{r=0}^{\alpha}\int_{\theta=0}^{\alpha\pi/2} e^{-r^2} \cdot r\,dr\,d\theta.$$

$$\Rightarrow \left(\Gamma\left(\tfrac{1}{2}\right)\right)^2 = 4\int_0^{\alpha} r\cdot e^{-r^2}\cdot dr \, \left|\theta\right|_0^{\pi/2}$$



$$\Rightarrow \left(\Gamma\left(\tfrac{1}{2}\right)\right)^2 = \frac{4\pi}{2}\int_{r=0}^{\alpha} e^{-r^2}\cdot r\, dr \longrightarrow (5)$$

Let $r^2 = t$

$$\Rightarrow 2r\,dr = dt \Rightarrow r\,dr = \frac{1}{2}dt$$

$$\therefore (5) \Rightarrow \left(\Gamma\left(\tfrac{1}{2}\right)\right)^2 = \cancel{2}\pi\int_{t=0}^{\alpha} e^{-t}\cdot \frac{1}{\cancel{2}}dt$$

$$\Rightarrow \left(\Gamma\left(\tfrac{1}{2}\right)\right)^2 = \pi\left|\frac{e^{-t}}{-1}\right|_0^{\alpha}$$

$$= \pi\left(-e^{-\alpha} - e^{0}\right)$$

$$\Rightarrow \left(\Gamma\left(\tfrac{1}{2}\right)\right)^2 = \pi(0+1)$$

$$\Rightarrow \Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}$$

5. $\beta(m,n) = \beta(n,m)$

$$\Gamma(1) = 1$$

$$\Gamma(n)\,\Gamma(1-n) = \frac{\pi}{\sin \pi}$$

$$\Gamma(n+1) = n\Gamma(n) \text{ or } n!$$

1. Show that $\int_0^{\pi/2} \sqrt{\sin\theta}\cdot d\theta \times \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin\theta}}\, d\theta = \pi$

Sol:- Let $I = \int_0^{\frac{\pi}{2}} \sqrt{\sin\theta}\cdot d\theta \times \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin\theta}}\, d\theta$

$$\Rightarrow I = I_1 \times I_2$$

$$\therefore I_1 = \int_0^{\pi/2} \sqrt{\sin\theta}\cdot d\theta$$

$$I_1 = \int_0^{\pi/2} \sin^{1/2}\theta \cdot d\theta$$

$$\Rightarrow I_1 = \int_0^{\pi/2} \sin^p\theta \cdot \cos^q\theta \cdot d\theta$$

$$\therefore p = \frac{1}{2},\ q = 0$$

$$\therefore I_1 = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$I_1 = \frac{1}{2}\beta\left(\frac{1/2+1}{2}, \frac{0+1}{2}\right)$$

$$I_1 = \frac{1}{2}\beta\left(\frac{3}{4}, \frac{1}{2}\right)$$

$$I_1 = \frac{\frac{1}{2}\,\Gamma\frac{3}{4}\,\Gamma\frac{1}{2}}{\Gamma\left(\frac{3}{4}+\frac{1}{2}\right)}$$

$$I_1 = \frac{\frac{1}{2}\,\Gamma\frac{3}{4}\,\Gamma\frac{1}{2}}{\Gamma\frac{5}{4}}$$

$$\therefore I = \frac{\cancel{2}\,\Gamma\frac{3}{4}\,\Gamma\frac{1}{2}}{\cancel{\Gamma\frac{1}{4}}} \cdot \frac{1}{\cancel{2}}\,\frac{\Gamma\frac{1}{4}\,\Gamma\frac{1}{2}}{\Gamma\frac{3}{4}}$$

$$\Rightarrow I = \left(\Gamma\tfrac{1}{2}\right)\left(\Gamma\tfrac{1}{2}\right)$$

$$\Rightarrow I = \sqrt{\pi}\cdot\sqrt{\pi}$$

$$\boxed{I = \pi}$$

2. Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{\tan\theta}\,.d\theta$

Sol:- $I = \int_0^{\frac{\pi}{2}} \frac{\sin^{1/2}\theta}{\cos^{1/2}\theta}\,d\theta$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \sin^{1/2}\theta.\cos^{-1/2}\theta.d\theta$$

$$\Rightarrow I = \int_0^{\pi/2} \sin^{p}\theta.\cos^{q}\theta.d\theta$$

$$\therefore p = \frac{1}{2},\ q = -1/2$$

$$\Rightarrow I = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$\Rightarrow I = \frac{1}{2}\beta\left(\frac{\frac{1}{2}+1}{2}, \frac{-1/2+1}{2}\right)$$

$$\Rightarrow I = \frac{1}{2}\beta\left(\frac{3}{4}, \frac{1}{4}\right)$$

$$\Rightarrow I = \frac{1}{2}\beta\,\frac{\Gamma\frac{3}{4}\cdot\Gamma\frac{1}{4}}{\Gamma\left(\frac{3}{4}+\frac{1}{4}\right)}$$

$$\Rightarrow I_1 = \frac{\frac{1}{2}\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{2}\right)}{\left(\frac{5}{4}-1\right)\Gamma\left(\frac{5}{4}-1\right)}$$

$$\Rightarrow I_1 = \frac{1}{2}\,\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{2}\right)}{\frac{1}{4}\Gamma\left(\frac{1}{4}\right)}$$

$$\Rightarrow I_1 = 2\,\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{4}\right)}$$

$$\therefore I_2 = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin\theta}}\,d\theta$$

$$\Rightarrow I_2 = \int_0^{\frac{\pi}{2}} \sin^{-1/2}\theta \, .\, \cos^{0}\theta \, .\, d\theta$$

$$\therefore p = -\frac{1}{2}, \quad q = 0$$

$$\therefore I_2 = \frac{1}{2}\beta\left(\frac{-\frac{1}{2}+1}{2}, \frac{0+1}{2}\right)$$

$$\Rightarrow I_2 = \frac{1}{2}\beta\left(\frac{1}{4}, \frac{1}{2}\right)$$

$$I_2 = \frac{1}{2}\,\frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{4}+\frac{1}{2}\right)}$$

$$I_2 = \frac{1}{2}\,\frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4}\right)}$$

$$\Rightarrow I = \frac{1}{2}\,\frac{\Gamma\frac{3}{4}\,\Gamma\frac{1}{4}}{\Gamma 1}$$

$$\Rightarrow I = \frac{1}{2}\,\Gamma\frac{3}{4}\,\Gamma\frac{1}{4}$$

$$\Rightarrow I = \frac{1}{2}\,\Gamma\left(1-\frac{1}{4}\right)\Gamma\frac{1}{4}$$

$$\Rightarrow I = \frac{1}{2}\cdot\frac{\pi}{\sin\left(\frac{\pi}{4}\right)}$$

$$\Rightarrow I = \frac{1}{2}\cdot\frac{\pi}{\frac{1}{\sqrt{2}}}$$

$$\Rightarrow I = \frac{\cancel{\sqrt{2}}\,\pi}{\sqrt{2}\,\cancel{\sqrt{2}}}$$

$$\boxed{I = \frac{\pi}{\sqrt{2}}}$$

3. Show that $\int_0^2 (4-x^2)^{3/2}\,dx = 3\pi$

Sol:- Let $I = \int_0^2 (4-x^2)^{3/2}\,dx \longrightarrow (1)$

Let $x = 2\sin\theta \Rightarrow \sin\theta = \frac{x}{2} \Rightarrow \theta = \sin^{-1}\left(\frac{x}{2}\right)$

$\Rightarrow dx = 2\cos\theta\,d\theta$

UL : $x = 2 \Rightarrow \theta = \frac{\pi}{2}$

LL : $x = 0 \Rightarrow \theta = 0$

$\therefore I = \int_0^{\pi/2} \left(4-4\sin^2\theta\right)^{3/2} \cdot 2\cos\theta\,d\theta$

$$\Rightarrow I = 2\int_0^{\frac{\pi}{2}} 4^{3/2}\left(\cos^2\theta\right)^{3/2}\cos\theta\, d\theta$$

$$\Rightarrow I = 2\int_0^{\pi/2} (2^{\cancel{2}})^{3/\cancel{2}} \cdot \cos^3\theta \cdot \cos\theta\, d\theta$$

$$\Rightarrow I = 2^4 \int_0^{\pi/2} \cos^4\theta\, d\theta$$

$$\Rightarrow I = 16\int_0^{\pi/2} \sin^0\theta \cdot \cos^4\theta \cdot d\theta$$

$$\therefore p = 0,\ q = 4$$

$$\therefore I = 16 \cdot \frac{1}{2}\beta\left(\frac{0+1}{2}, \frac{4+1}{2}\right)$$

$$\Rightarrow I = 8\beta\left(\frac{1}{2}, \frac{5}{2}\right)$$

$$\Rightarrow I = \frac{8\,\Gamma\left(\frac{1}{2}\right)\cdot\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{1}{2}+\frac{5}{2}\right)}$$

$$\Rightarrow I = \frac{8\,\Gamma\left(\frac{1}{2}\right)\cdot\Gamma\left(\frac{5}{2}\right)}{\Gamma(3)}$$

$$\Rightarrow I = \frac{8\,\Gamma\left(\frac{1}{2}\right)\left(\frac{5}{2}-1\right)\Gamma\left(\frac{5}{2}-1\right)}{(3-1)\,\Gamma(3-1)}$$

$$\Rightarrow I = \frac{8\,\Gamma\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\Gamma\left(\frac{3}{2}\right)}{2\,\Gamma(2)}$$

$$\Rightarrow I = \frac{8\,\Gamma\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{2.1\,\Gamma 1}$$

$$\Rightarrow I = \frac{\overset{2}{\cancel{8}} \times \frac{3}{\cancel{2}} \times \frac{1}{\cancel{2}} \times \Gamma\frac{1}{2} \times \Gamma\frac{1}{2}}{\cancel{2}}$$

$$\Rightarrow I = 3\sqrt{\pi}\,\sqrt{\pi}$$

$$\boxed{I = 3\pi}$$

4. Evaluate $\int_0^1 x^{3/2}(1-x)^{1/2}\,dx$

Sol:- Let $x = \sin^2\theta$, $dx = 2\sin\theta\cos\theta\,d\theta$

$$\sin\theta = \sqrt{x}$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{x}}\right)$$

UL. $x = 1$, $\theta = \pi/2$

LL. $x = 0$, $\theta = 0$

$$I = \int_0^{\pi/2} (\sin^2\theta)^{3/2}\,(1-\sin^2\theta)^{1/2} .\, 2\sin\theta\cos\theta\,d\theta$$

$$\Rightarrow I = 2\int_0^{\pi/2} \sin^3\theta . \cos\theta \sin\theta . \cos\theta . d\theta$$

$$\Rightarrow I = 2\int_0^{\pi/2} \sin^4\theta . \cos^2\theta . d\theta$$

$\therefore p = 4$, $q = 2$

$$I = \cancel{2} . \frac{1}{\cancel{2}}\,\beta\left(\frac{4+1}{2}, \frac{2+1}{2}\right)$$

$$\Rightarrow I = \beta\left(\frac{5}{2}, \frac{3}{2}\right)$$

$$\Rightarrow I = \beta \; \frac{\Gamma\left(\frac{5}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{5}{2}+\frac{3}{2}\right)}$$

$$\Rightarrow I = \frac{\Gamma\left(\frac{5}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma(4)}$$

$$\Rightarrow I = \frac{\left(\frac{3}{2}\right)\Gamma\left(\frac{3}{2}\right) \cdot \frac{1}{2}\left(\Gamma\left(\frac{1}{2}\right)\right)}{3\Gamma(3)}$$

$$\Rightarrow I = \frac{\frac{3}{2}\left(\frac{1}{2}\right)\left(\Gamma\left(\frac{1}{2}\right)\right)\frac{1}{2}\left(\Gamma\left(\frac{1}{2}\right)\right)}{3 \cdot 2 \cdot 1 \cdot \Gamma(1)}$$

$$\Rightarrow I = \frac{\frac{\cancel{3}}{2} \times \frac{1}{2} \times \frac{1}{2} \times \Gamma\left(\frac{1}{2}\right) \times \Gamma\left(\frac{1}{2}\right)}{\cancel{6}\,2}$$

$$\Rightarrow I = \frac{1}{2^4}\sqrt{\pi}\sqrt{\pi}$$

$$\boxed{I = \frac{\pi}{16}}$$

5. Show that $\int_0^{\infty} \frac{e^{-x^2}}{\sqrt{x}} dx \times \int_0^{\infty} \sqrt{x}\, e^{-x^2} dx = \frac{\pi}{2\sqrt{2}}$

Sol:- Let $I = \int_0^{\infty} \frac{e^{-x^2}}{\sqrt{x}} dx \times \int_0^{\infty} \sqrt{x} \cdot e^{-x^2} \cdot dx$

$$I = I_1 \times I_2$$

$$I_1 = \int_0^{\infty} \frac{e^{-x^2}}{\sqrt{x}} dx$$

$$\Rightarrow I_1 = \int_0^{\infty} e^{-x^2} \cdot x^{-1/2} dx$$

$$\Rightarrow I_1 = \int_0^{\infty} e^{-x^2} \cdot x^{2n-1} dx$$

$$2n-1 = -1/2$$

$$2n = 1/2$$

$$n = \frac{1}{4}$$

$$I_1 = \frac{\Gamma n}{2}$$

$$\boxed{I_1 = \frac{1}{2} \Gamma \frac{1}{4}}$$

$$I_2 = \int_0^{\infty} e^{-x^2} \cdot x^{1/2} dx$$

$$I_2 = \int_0^{\infty} e^{-x^2} \cdot x^{2n-1} dx$$

$$\therefore\ 2n-1 = \frac{1}{2}$$

$$\Rightarrow 2n = \frac{3}{2}$$

$$\Rightarrow n = \frac{3}{4}$$

$$\therefore\ I_2 = \frac{\Gamma(n)}{2}$$

$$\boxed{I_2 = \frac{1}{2}\Gamma\left(\frac{3}{4}\right)}$$

$$I = I_1 \times I_2$$

$$\Rightarrow I = \frac{1}{2}\Gamma\left(\frac{1}{4}\right)\frac{1}{2}\Gamma\left(\frac{3}{4}\right)$$

$$\Rightarrow I = \frac{1}{4}\Gamma\left(\frac{1}{4}\right)\Gamma\left(1-\frac{1}{4}\right)$$

$$\Rightarrow I = \frac{1}{4}\frac{\pi}{\sin\left(\frac{\pi}{4}\right)}$$

$$\Rightarrow I = \frac{1}{4}\cdot\sqrt{2}\,\pi$$

$$\Rightarrow I = \frac{\cancel{\sqrt{2}}\ \pi}{2\times\sqrt{2}\cdot\cancel{\sqrt{2}}}$$

$$\Rightarrow I = \frac{\pi}{2\sqrt{2}}$$